

ABDULRAHMAN REDA ABUZAID

Full-Stack Developer | Flutter & API Specialist | Embedded Systems & IoT Enthusiast

 6th of October City, Giza, Egypt |  abdoreda2008gtr@gmail.com | 
+201507064713  [LinkedIn](#) |  [GitHub/Portfolio](#)

PROFESSIONAL SUMMARY

Highly motivated and technically versatile developer with a robust foundation in **Full-Stack Development** (Python, JavaScript, C#, ASP.NET, React.js, Node.js) and **Enterprise IT Systems**. Recently specialized in **Flutter** mobile development, **Embedded Systems (STM)**, and **IoT**. Proven ability to deliver practical solutions, from configuring LAN networks and managing MySQL/PostgreSQL databases to developing advanced AI-driven research projects. Maintains an active GitHub portfolio with 6+ public repositories. Seeking a challenging role to drive innovation and apply a diverse skill set in a professional technology environment.

TECHNICAL SKILLS

- **Programming:** Python, JavaScript, C#, C, C++, PHP, Java, Bash, Dart, HTML5, CSS3.
- **Frameworks & Libraries:** Flutter, React.js, Node.js, Express.js, ASP.NET, Flask, WPF, Bootstrap, Tailwind CSS.
- **Embedded & IoT:** Embedded Systems (STM), IoT, Arduino IDE, MPU6050, ESP32, Raspberry Pi, Sipeed Lichee RV Nano.
- **APIs & Databases:** RESTful API Development & Integration, Postman, MySQL, PostgreSQL, MongoDB, Firebase, Hive, SQL.
- **ERP & Tools:** Odoo (Technical Foundation, Configuration), VS Code, PyCharm.

- **Version Control:** Git, GitHub (Branching, Commits, Pull Requests, Collaboration).
 - **IT & Networking:** Switch Configuration, LAN Systems, Printer Setup, System Troubleshooting.
 - **Design/UX:** Figma, Adobe XD, Webflow, Wireframing, User Flow Design, Responsive Design, Pose Estimation.
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EXPERIENCE & PROJECTS

Capstone Project: MedReminder – Family-Based Smart Medication System

- **Description:** A smart mobile application designed to organize medication schedules and track user adherence, linking family members to ensure no doses are missed. Features an offline-first system with smart escalation logic.
- **Technologies:** Flutter, Dart, Provider (State Management), Hive (Local Storage), Flask (Python), REST API, flutter_local_notifications.
- **Key Achievements:**
 - Developed a cross-platform Flutter app for medication management and dosage alerts.
 - Implemented an advanced notification system that works offline with precise timing.
 - Built a core family-linking feature allowing users to monitor medication adherence of family members.
 - Designed a smart escalation system to notify family members via app or email if doses are ignored.
 - Ensured secure data storage using `flutter_secure_storage` and unit/widget testing for stability.

Technical Project: POS System Installation & Network Configuration

- **Al-Matam Al-Arabi | Nasr City, Egypt | Project-Based**
 - Engineered and implemented a complete cashier (POS) system using a local server setup.

- Configured and integrated 8 network printers via a dedicated switch, optimizing LAN communication.
- Managed database integration by setting up a local MySQL server for secure billing and order processing.
- Executed system troubleshooting and device integration under real-world operational constraints.

Co-Founder & Instructor

- **Nova Technology | Giza, Egypt | July 2024 – Present**
 - Co-founded a youth tech education initiative, merging technical development with instructional design.
 - Designed beginner-level programming curricula (C++ Basics, Scratch) for delivery in Egypt and the UAE.
 - Mentored students through interactive, project-based methods to ensure grasp of core concepts.
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RESEARCH PROJECTS (INDEPENDENT)

Note: These projects explored advanced concepts subsequently implemented by industry leaders such as Meta and Samsung.

Pulse – AI Emotion Layer System

- **Description:** A system adding an “artificial emotion layer” to AI, enabling stateful, adaptive behavior based on an internal “Emotion Vector” (E).
- **Key Contributions:** Designed the Emotion Engine and Heart Core concepts; developed frameworks for input appraisal, experience memory, and interoception; modeled how emotions modulate decision-making (Policy Modulation).

Echoo – AI Communication Platform

- **Description:** An advanced real-time translation platform breaking communication barriers (language, hearing, emotion) with human-like

interaction.

- **Key Contributions:** Designed a “Bi-Mode” system for voice/emotion imitation; developed a secure biometric identity system for voice cloning prevention; engineered camera-based sign language translation using Pose Estimation.

Akuma – Smart AI Glasses

- **Description:** A smart glasses concept integrating a camera, AI/OpenCV for visual understanding, and voice interaction.
- **Key Contributions:** Modeled a system for real-time object recognition and user feedback; researched hardware constraints (ESP32 vs. Raspberry Pi); proposed advanced features like Eye Tracking and brain-signal integration.

COA – Company of AI's

- **Description:** A virtual company composed of specialized AI agents (UI/UX, Frontend, Backend, PM) collaborating to complete complex projects rapidly.
 - **Key Contributions:** Developed the agent collaboration workflow and task distribution logic; proposed using Analytic Network Process (ANP) for autonomous decision-making and prioritization.
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EDUCATION & ACHIEVEMENTS

- **High School Diploma – Applied Technology Track**
 - Eva International Applied Technology School (IATS) | Expected Graduation: 2026
 - Academic Standing: 3.1 GPA.
 - Focus: Engineering, Computer Science, and Technical Innovation.
- **Awards:**
 - **Galactic Problem Solver Award** (NASA International Space Apps Challenge 2024).
 - **Certificate of Appreciation** (NASA Space Apps Cairo Hackathon 2024).
- **Certifications:**

- Responsive Web Design – Freecodecamp (2024).
 - DOM Fundamentals Certification – Satr.codes (2025).
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ADDITIONAL SKILLS

- **Soft Skills:** Leadership, Team Collaboration, Time Management, Self-Learning, Creativity, Public Speaking, Teaching & Mentoring, Technical Documentation, System Analysis.
- **Languages:** Arabic (Native), English (Upper-Intermediate – B2), Spanish (Basic).